

ECEN 662: Estimation & Detection Theory

Lecture 19: Markov Chain Monte Carlo Sampling

Tie Liu

Texas A&M University

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 - The samples *converge* in distribution to the target distribution.
 - We can construct a *consistent* estimator for expectations, even though the samples are *not* independent.

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- **Continuous Case:** If P has a *transition density* $p(x, y)$, π is stationary if it satisfies the integral equation:

$$\int_{\mathcal{X}} \pi(x) p(x, y) dx = \pi(y)$$

Fundamental Theorem of Markov Chains

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 - **Ergodic Theorem:** Under mild conditions on h , the sample average converges almost surely to the expectation:

$$\frac{1}{T} \sum_{t=0}^{T-1} h(X_t) \xrightarrow{a.s.} \mathbb{E}_{\pi}[h(X)] \quad \text{as } T \rightarrow \infty$$

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- **Implication:** If direct sampling from π is intractable, we can simulate a Markov chain that has π as its stationary distribution.
- *Note:* We assume these technical conditions hold for the algorithms discussed herein.

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- **Detailed Balance Condition:** If P has a *density* $p(x, y)$, reversibility is equivalent to the *detailed balance condition*:

$$\pi(x)p(x, y) = \pi(y)p(y, x) \quad \forall x, y \in \mathcal{X}$$

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- **Proof of Stationarity:** Setting $B = \mathcal{X}$ (the entire state space) in the equation above gives:

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- **Conclusion:** $X_0 \sim \pi \implies X_1 \sim \pi$. Reversibility is a sufficient condition for stationarity.

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- **Examples:**
 - *Continuous Space* ($\mathcal{X} = \mathbb{R}^d$): $Q(x, x') = \mathcal{N}(x, \sigma^2 I)$.
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- **Requirement:** For each x , we must be able to efficiently *sample* from $Q(x, \cdot)$.

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- **Goal:** Choose $\alpha(x, y)$ such that the resulting Markov chain is *reversible* with respect to π .

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- **Combined Transition Kernel:**

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and the corresponding transition density is:

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Choosing Acceptance Probability α

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- **Key Insight:** The second term in $p(x, y)$ is only non-zero when $x = y$. For $x = y$, detailed balance trivially holds. Therefore, for $x \neq y$, it suffices to pick $\alpha(x, y)$ to satisfy:

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- **Constraint:** Because α is an acceptance probability, we must ensure that $\alpha(x, y) \in [0, 1]$ for all x, y .

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- The subscript B in the second choice stands for Barker. While both yield the correct stationary distribution, the *Metropolis-Hastings* choice (α^*) is the standard in practice.

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 - The proposal terms cancel out perfectly:

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- **Independent Proposal:**
 - If the proposal does not depend on the current state, i.e., $q(x, y) = q(y)$.
 - The acceptance probability becomes a ratio of importance weights:

$$\alpha(x, y) = \min \left(1, \frac{\pi(y)q(x)}{\pi(x)q(y)} \right) = \min \left(1, \frac{w(y)}{w(x)} \right)$$

where $w(\cdot) = \pi(\cdot)/q(\cdot)$.

Gibbs Sampling: A Powerful Special Case

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- **The Algorithm:** At iteration $t + 1$, we sweep through all d coordinates sequentially:

$$\begin{aligned}x_1^{(t+1)} &\sim \pi(x_1 \mid x_2^{(t)}, x_3^{(t)}, \dots, x_d^{(t)}) \\x_2^{(t+1)} &\sim \pi(x_2 \mid x_1^{(t+1)}, x_3^{(t)}, \dots, x_d^{(t)}) \\&\vdots \\x_d^{(t+1)} &\sim \pi(x_d \mid x_1^{(t+1)}, x_2^{(t+1)}, \dots, x_{d-1}^{(t+1)})\end{aligned}$$

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- **Key Property:** Gibbs sampling is equivalent to MH using the full conditionals as the proposal distributions. Remarkably, this guarantees the acceptance probability is *exactly 1* for every step.

Asymptotic Variance

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- **Definition:** We define the *asymptotic variance* of $\hat{h}_T$ as:

$$\sigma_h^2(P) := \lim_{T \rightarrow \infty} \left(T \text{Var}(\hat{h}_T) \right)$$

which is independent of the distribution of X_0 and is the standard metric used to compare different Markov chains (kernels P) that have the *same* stationary distribution.

Peskun ordering of Markov chains

- (*Peskun-Tierney theorem*) Let P_1 and P_2 be two transition kernels, both *reversible* with respect to π . We have $\sigma_h^2(P_1) \leq \sigma_h^2(P_2)$ if

$$P_1(x, B \setminus \{x\}) \geq P_2(x, B \setminus \{x\})$$

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Optimality of the Metropolis-Hastings Ratio

- **Claim:** For any acceptance probability $\alpha(x, y)$ satisfying the *detailed balance condition*:

$$\pi(x)\alpha(x, y)q(x, y) = \pi(y)\alpha(y, x)q(y, x)$$

we have:

$$P_{\alpha^*}(x, B \setminus \{x\}) \geq P_{\alpha}(x, B \setminus \{x\})$$

for any $x \in \mathcal{X}$ and $B \in \mathcal{B}(\mathcal{X})$, where α^* is the standard Metropolis-Hastings choice:

$$\alpha^*(x, y) = \min \left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \right)$$

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- **Implication:** The standard Metropolis-Hastings acceptance probability α^* guarantees the highest probability of moving to a new state among *all* reversible algorithms with the same proposal Q . By Peskun's Theorem, it therefore minimizes the asymptotic variance!

Proof of Optimality: Setup

- To prove $P_{\alpha^*}(x, B \setminus \{x\}) \geq P_{\alpha}(x, B \setminus \{x\})$, it suffices to show that the continuous part of the transition density satisfies:

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- Because $\alpha^*(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right)$, the natural way to proceed is to split the proof into two cases based on the ratio:

$$\frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \geq 1 \quad \text{and} \quad \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} < 1.$$

Proof of Optimality: Analyzing the Cases

- **Case 1:** $\frac{\pi(y)q(y,x)}{\pi(x)q(x,y)} \geq 1$. In this case, we have

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- **Case 2:** $\frac{\pi(y)q(y,x)}{\pi(x)q(x,y)} < 1$. In this case, we have

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and hence,

$$p_{\alpha^*}(x, y) = \alpha^*(x, y)q(x, y) = \frac{\pi(y)q(y, x)}{\pi(x)}$$

- To compare this to $p_\alpha(x, y)$, we use the *detailed balance condition* for α :

$$p_\alpha(x, y) = \frac{\pi(y)\alpha(y, x)q(y, x)}{\pi(x)}$$

Since $\alpha(y, x) \leq 1$, it immediately follows that:

$$p_\alpha(x, y) \leq \frac{\pi(y)q(y, x)}{\pi(x)} = p_{\alpha^*}(x, y)$$

completing the proof!

Autocorrelation and Effective Sample Size

- To understand why the asymptotic variance $\sigma_h^2(P)$ differs from the i.i.d. case, we expand the variance of our estimator:

$$\text{Var}(\hat{h}_T) = \frac{1}{T^2} \sum_{t=0}^{T-1} \text{Var}(h(X_t)) + \frac{2}{T^2} \sum_{0 \leq s < t \leq T-1} \text{Cov}(h(X_s), h(X_t))$$

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- The covariance terms exist because MCMC samples are *correlated*. We quantify this using the *integrated autocorrelation time* (IACT), denoted as τ :

$$\tau = 1 + 2 \sum_{k=1}^{\infty} \rho(k)$$

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- **Effective Sample Size (ESS):** We define $\text{ESS} = T/\tau$. This represents the equivalent number of *independent* samples that would yield the same estimation variance as our T correlated MCMC samples.

Convergence Diagnostics: Burn-in and Mixing

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- **Visual Diagnostics:** We often use *trace plots* (plotting X_t versus t) to assess both burn-in (transient phase) and mixing (stationarity and exploration).

Lecture Summary: Markov Chain Monte Carlo

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- **Powerful Variants:** *Special Cases* like the original Symmetric Metropolis algorithm and *Gibbs Sampling* (updating via full conditionals) provide highly efficient, targeted tools for high-dimensional problems.
- **Practical Evaluation:** Unlike i.i.d. sampling, MCMC estimators suffer from initialization bias and correlation. We must discard early transient samples (*burn-in*) and monitor the *Effective Sample Size (ESS)* to evaluate chain mixing.